Part H — Computer vision & OCR

ml-ladder writeup · part H

2026-09-24

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What part H settled **28**

Status: complete (H1–H6, plus the real-model runs) · **Constraint:** synthetic data with exact ground truth, rendered offline; small budgets
Numbers: seeds 0–2, reference seed 0 · raw values in `data/h-metrics.json`
Reproduce: `ladder run H1` … `H6` (needs `uv sync --extra torch --extra ocr`), or `uv run python tools/writeup_figures.py H`
Hardware: H1–H5 swept on a Colab T4; H6 is arithmetic and runs anywhere

Every part before this one was handed its metric. Accuracy needs no argument — the prediction equals the label or it does not — and even part D's probes were scored by counting correct answers. Detection has no such luxury. A model emits a variable number of boxes, each with a score and a class, and *nothing* says which prediction corresponds to which object. The metric has to be built, and building it involves choices that change the answer.

So part H is about **interfaces**: between a prediction and the thing it is supposed to be a prediction *of*, and between one stage of a pipeline and the next. H1 constructs a metric from nothing. H2 discovers that averaging over the wrong pixels hides the effect it is looking for. H3 and H4 find that the interface between an image and a string can be drawn two ways with opposite failure modes. H5 measures a real two-stage pipeline and finds **99.98% of its error living in the interface between the stages**. H6 prices what it costs to delete that interface entirely, which is what the field actually did.

The data is rendered rather than downloaded, in `src/mlladder/synth.py`. Attributing a pipeline's error to a stage needs ground truth that is exactly right — which box, which character, which pixel — and real OCR corpora come with annotation noise that is itself a research topic. The fonts come from matplotlib's bundled TTFs rather than the system's, because otherwise every number here would be a property of the machine that produced it. That reproducibility is verified, not assumed: the rendered pixels are byte-identical across matplotlib 3.10/3.11 and Pillow 11.3/12.3, checked by hashing the output on two machines and now guarded by a test.

Notation: boxes are (x_1, y_1, x_2, y_2) in pixels, IoU is intersection over union, CER and WER are edit distances normalized by reference characters and words, and T is a sequence length in image columns or tokens.

H1 — Detection, and the metric as a construction

Question. Detection's score depends on three choices that have nothing to do with the detector: what overlap counts as a match, how duplicates are removed, and how a precision-recall curve becomes one number. How much do those choices move the answer?

The idea

Object detection means finding every object in an image and drawing a box around each, with a class label and a confidence score. Unlike classification, the number of answers varies per image.

This rung builds a **YOLO-style** detector ("you only look once"): split the image into an 8×8 grid, and have every cell predict, in a single forward pass, (1) whether an object's centre lies in it, (2) the box, as an offset within the cell plus a width and height, and (3) the object's class. One convolutional network produces all 64 predictions at once. Neighbouring cells often fire for the same object, so **non-maximum suppression** (**NMS**) removes duplicates: keep the most confident box, delete any other box that overlaps it too much, repeat.

The harder problem is *scoring* it, because no single rule says which prediction belongs to which object. The standard score, **mean average precision (mAP)**, is built from three choices, and this rung measures how much each one moves the answer.

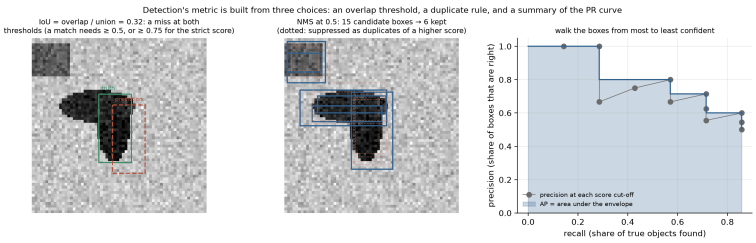


Figure 1: IoU, NMS and average precision

Left: IoU, the overlap measure every detection metric is built on (this pair overlaps too little to count). Middle: a cluster of candidate boxes before and after NMS, using the rung's own functions. Right: ranking predictions by confidence traces a precision-recall curve, and AP is the area under its upper envelope.

The math

IoU is the whole foundation, and it is one ratio:

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

The intersection of two axis-aligned boxes is a box — the coordinate-wise max of the near corners and min of the far ones — which is why this is cheap enough to evaluate on every prediction-truth pair.

Non-max suppression is three lines and one assumption: that two boxes overlapping by more than τ_{nms} describe the same object. Sort by score, keep the best, discard everything it overlaps, repeat. The assumption is false whenever two real objects overlap that much, which is why the threshold is a dial rather than a constant.

Average precision walks the predictions in descending score order. A prediction is a true positive if it clears τ_{iou} against a truth in the same image *that nothing better has already claimed* — the claiming rule is what makes a duplicate cost something, and therefore what makes NMS worth running at all. Each prefix gives a (r, p) point, and AP is the area under the monotone envelope:

$$\mathrm{AP} = \sum_k (r_k - r_{k-1}) \max_{\tilde{r} \geq r_k} p(\tilde{r}).$$

Taking the running maximum of precision from the right is the "interpolated" part. It matters more than it looks: a duplicate that lands *after* full recall has been reached costs nothing at all under the envelope, so NMS only pays where duplicates outrank genuine detections elsewhere.

mAP averages AP over classes — over the classes that are *present*. Scoring an absent class 0 rather than skipping it is a silent way to make a detector look bad on a dataset that never showed it that class.

What we measure, and why

- **IoU** (**intersection over union**) of a predicted and a true box: the area they share divided by the area they cover together. 1 is a perfect box, 0 no overlap. A prediction *counts* as correct only above a threshold.
- **Precision** — the fraction of predicted boxes that are correct. **Recall** — the fraction of true objects that were found. Lowering the confidence cut-off trades precision for recall.
- **AP** (**average precision**) — the area under the precision-recall curve for one class. **mAP** — its mean over classes. **mAP@0.5** accepts a box at IoU ≥ 0.5 (a loose fit). **mAP@0.75** demands a tight one. Reporting both shows whether errors come from *finding* objects or from *placing* the box.
- **Boxes per image vs NMS threshold** — against the true 2.0 objects per image, this shows whether NMS is removing duplicates or real objects.
- **Matched IoU distribution** — how tight the boxes are, which explains the gap between the two mAP numbers.

What I built

A YOLO-style detector with one prediction per cell of an 8×8 grid on 64×64 scenes: objectness, a box, and a class distribution per cell. The box is parameterised relative to its cell — centre as a sigmoid offset inside it, size as a log-ratio of the image — because predicting absolute coordinates asks every cell to learn where it is, while offsets let one shared convolution work everywhere. Ground-truth boxes are assigned to the cell holding their centre, which caps recall when two centres share a cell; that happens to 0.17% of objects here and is reported rather than left implicit.

Results

Setup: 3000 train / 600 test scenes, 12 epochs; medians over seeds 0–2.

NMS threshold	boxes/image	mAP@0.5	mAP@0.75
none	4.30	0.847	0.652
0.1	1.91	0.923	0.634
0.3	1.99	0.953	0.649
0.5	2.20	**0.955**	0.650
0.7	3.38	0.908	**0.672**
0.9	4.30	0.847	0.652

Ground truth averages 2.0 objects per image, which NMS at 0.3 reproduces almost exactly (1.99). `best/nms_threshold` is 0.5 on every seed.

What surprised me

The two IoU thresholds disagree about what the NMS dial is *for*. Across the sweep, mAP@0.5 moves by **0.109** and mAP@0.75 by only **0.032** — and at 0.75 the best threshold is 0.7, not 0.5. At the strict threshold, duplicate removal stops mattering because localization has become the binding constraint, and a knob that looks important at one threshold is nearly inert at the other.

The localization penalty itself is the largest number in the rung: **0.955 at IoU 0.5 against 0.650 at 0.75**, a drop of 0.316 on *unchanged predictions*. I chased that rather than accept it, and the chase is most of what I learned here. It is not a grid bug — the predicted centre is unbiased to 0.1 px,

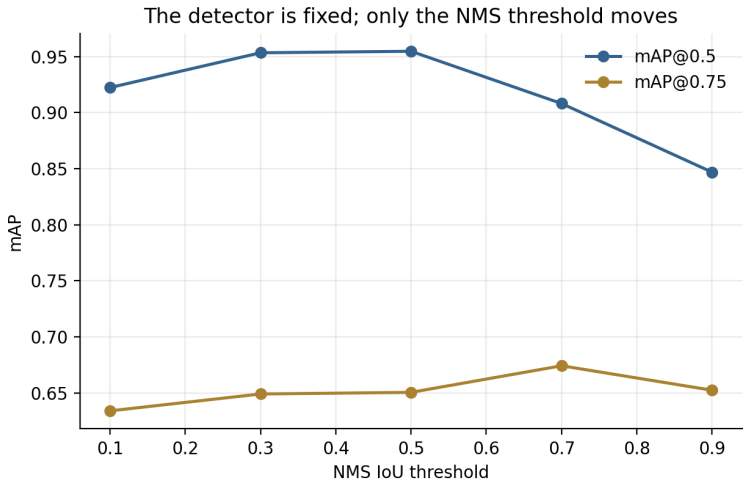


Figure 2: NMS sweep

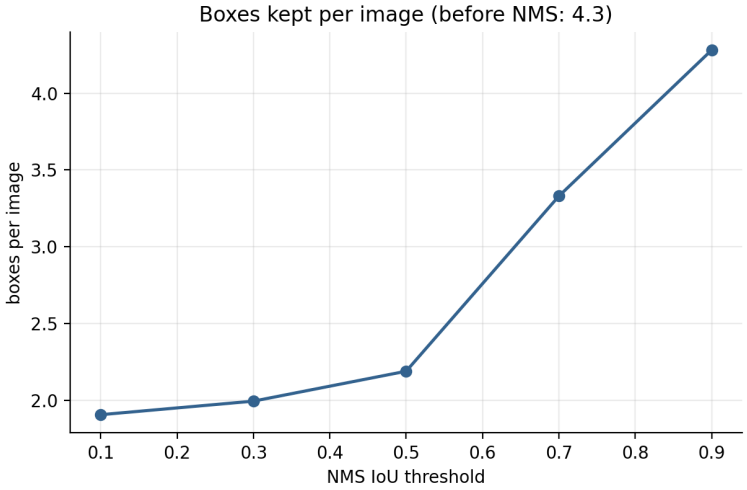


Figure 3: Boxes per image

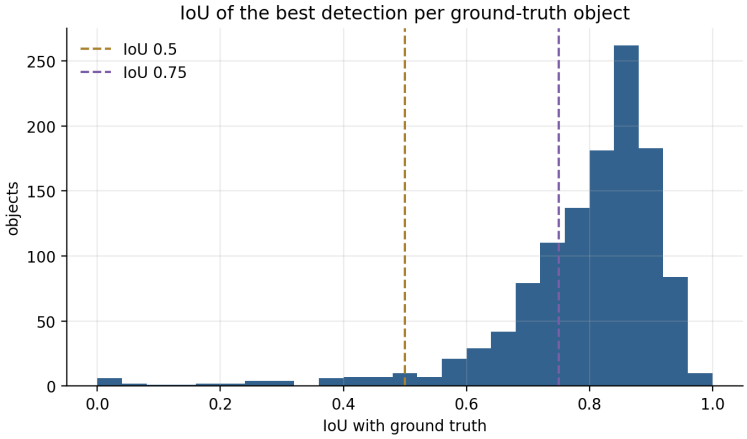


Figure 4: Matched IoU

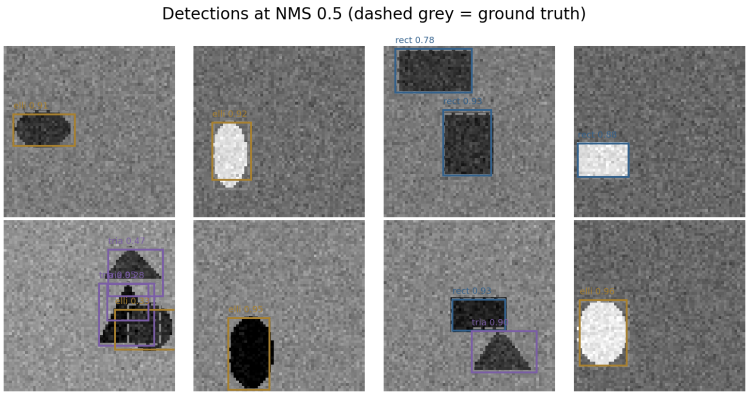


Figure 5: Detections

and feeding the encoder's own targets through the decoder round-trips to IoU > 0.99, which is now a test. It is not shape fill either: my hypothesis was that a triangle, filling half its box, would localize worst, but the width bias is identical across rectangle, ellipse and triangle (-1.46, -1.44, -1.49 px). And it is not Jensen's inequality from the log-size parameterisation, which accounts for only ~0.6 px of it. The residual is biased in log space itself (-0.067), where the *variance* (std 0.11, an 11% relative size error) dominates the bias anyway. That variance is what caps matched IoU at 0.792 mean, with 97% of objects above 0.5 but only 11% above 0.9.

Per class at the strict threshold the ordering is ellipse $0.698 >$ rectangle $0.629 >$ triangle 0.579, which my fill hypothesis gets exactly half right: the triangle is worst, but a rectangle *is* its box and still loses to an ellipse. I do not have an explanation I believe, and one sentence of honest ignorance beats a plausible story.

Open threads

Anchors with aspect priors, which is the standard answer to a size regression this noisy. Focal loss for the 30:1 negative imbalance. And DETR's set prediction, which removes NMS from the problem entirely rather than tuning it — the same move H6 makes one level up.

Further reading

YOLO; Faster R-CNN; Focal Loss; DETR.

H2 — Segmentation, and averaging over the wrong pixels

Question. Skip connections are supposed to restore the fine detail that pooling destroys. How much are they worth, and where?

The idea

Semantic segmentation labels every pixel: this pixel is background, that one is part of a triangle. The output is an image-sized map of classes.

The standard architecture is the **U-Net**. An *encoder* repeatedly convolves and pools, shrinking the image while building up context: by the bottom it knows *what* is in the image, but at coarse resolution. A *decoder* upsamples back to full size. The trouble is that pooling threw away exactly

the fine spatial detail needed to place edges precisely. U-Net's fix is **skip connections**: at each resolution the decoder concatenates the encoder's feature map from the same level, handing the fine detail straight across.

The rung ablates the skips and asks what they are worth, and discovers the answer depends on *which pixels* are averaged over.

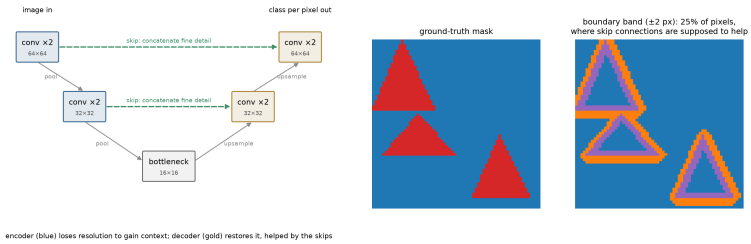


Figure 6: U-Net and the boundary band

Left: the U shape, with the skips (green) crossing from encoder to decoder at matching resolutions. Right: a ground-truth mask, and its boundary band, the pixels within 2 px of an edge. Skips are supposed to matter there, and almost nowhere else.

The math

The U-Net's claim is architectural. An encoder that pools discards high frequencies; upsampling cannot invent them back. A skip connection concatenates the pre-pool features into the decoder, giving the signal a route around the bottleneck. It is B4's residual argument in a second setting — the same idea one level up from a block.

Dice against cross-entropy is a question about weighting. Cross-entropy averages over pixels, so a class owning 3% of the image contributes 3% of the gradient. Soft Dice normalizes per class:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{C}\sum_{c} \frac{2\sum_i p_{ic} y_{ic} + \epsilon}{\sum_i p_{ic} + \sum_i y_{ic} + \epsilon}$$

Each class contributes one term regardless of its pixel count. The hard Dice score uses an argmax and has zero gradient; replacing the indicator with the softmax probability p_{ic} is what makes it trainable. Since the *metric* (IoU per class) is itself area-normalized, the two objectives genuinely disagree about what "good" means on a background that owns 86% of the pixels.

IoU from the confusion matrix is the diagonal over the row plus column minus the double-counted diagonal, and Dice and IoU are monotone in each other, $D = 2I/(1+I)$ — which the tests use to check one against the other.

What we measure, and why

- **Per-class IoU** — for each class, the overlap between predicted and true pixel sets divided by their union. **mIoU** averages it over classes. Accuracy would be misleading here: 86% of pixels are background, so predicting "background" everywhere scores 0.86.
- **Foreground mIoU** — mIoU over the three shape classes only, so the easy background class cannot inflate it.
- **Boundary mIoU** — the same score restricted to the 2-px band around true edges. This is where skip connections should act, and averaging over the whole image dilutes the effect with interior pixels a coarse model already gets right.

- **Loss arms** (cross-entropy, Dice, both) — Dice directly optimises overlap and is the standard remedy for class imbalance. It is tested rather than assumed.

What I built

A two-level U-Net on the same scenes H1 detects in. The no-skip arm keeps its entire decoder and loses only the concatenated channels, so the comparison is not confounded by capacity: 66,292 parameters against 59,812. Plus a boundary band — pixels within 2px of a label change, computed without scipy by disagreeing with a shifted copy and dilating — covering 11.7% of pixels.

Results

(2000 train / 500 test, 10 epochs; medians over seeds 0–2). Pixel shares: background 0.856, rectangle 0.063, ellipse 0.048, triangle 0.032.

arm	foreground mIoU	boundary mIoU
skips + cross-entropy	0.9495	0.9237
no skips + cross-entropy	0.9378	0.8921
skips + Dice	**0.9573**	**0.9325**
skips + CE + Dice	0.9572	0.9327

	whole image	boundary band
skip gain	+0.0129	**+0.0299**

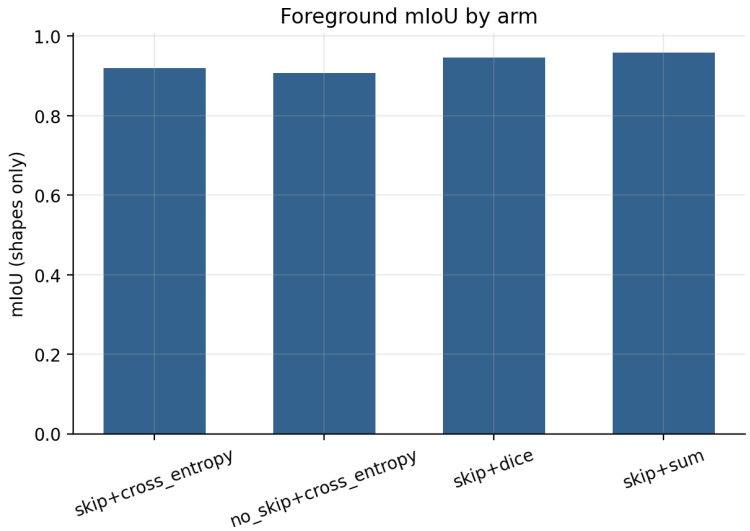


Figure 7: Arms

What surprised me

The first pass made the ablation look boring: +0.013 foreground mIoU for skips, everything above 0.94, a ceiling rather than a result. Then I scored the mechanism instead of the average. The argument for skip connections is about high frequencies — edges — and whole-image mIoU is dominated by object interiors, where a roughly-right blob already scores well. Restricting the

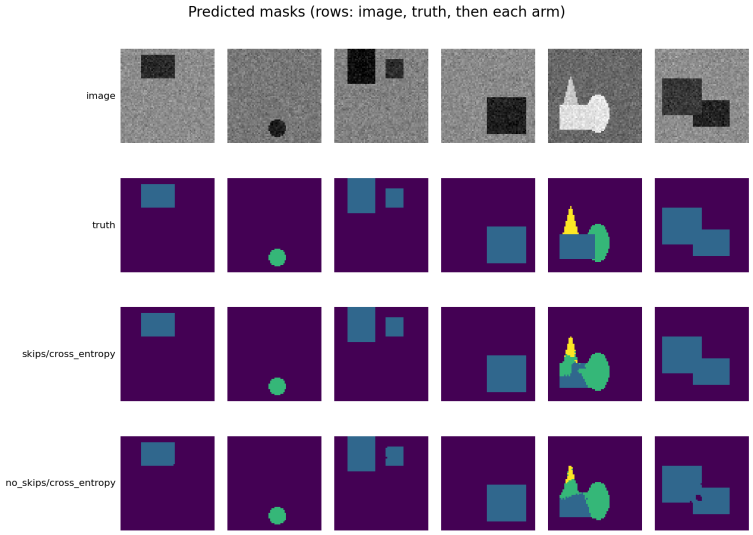


Figure 8: Masks

same predictions to a 2px band around each true boundary gives **+0.030 against +0.013, 2.3× larger**, on all three seeds. The effect was always there; the metric was averaging the place skips act together with the place they don't.

And a claim that did not survive the sweep. On a single seed I recorded that Dice beats cross-entropy, which looked like the textbook class-imbalance story holding. Across three seeds `best/loss_is_dice` is $[0, 1]$ — Dice wins twice and loses once — over a loss spread of 0.012. Dice and CE+Dice are statistically indistinguishable here (0.9573 against 0.9572). I had reported the single-seed version as a finding before the sweep, and it was noise.

Open threads

A deeper U-Net, where more pooling should widen the skip gap. Thin structures rather than solid shapes, which is the regime the boundary result predicts skips matter most in. And boundary-weighted losses, which optimize directly for the band this rung had to invent to see the effect.

Further reading

U-Net; Mask R-CNN; Segment Anything.

H3 — CTC, and what "held out" actually held out

Question. Reading text off an image has no alignment: a 128-pixel line holds an unknown number of characters at unknown positions. How much does removing that requirement cost, and where does the remaining error live?

The idea

Text recognition (**OCR**) turns an image of a line of text into a string. The difficulty is **alignment**: nobody says which pixel columns belong to which character, and the number of characters varies.

A **CRNN** (convolutional recurrent network) reads the line as a sequence of vertical strips: convolutions squash the image's height to 1 and keep 32 columns across its width, and a bidirectional LSTM (C1) passes context along the row. Every column then predicts a character *or a special blank symbol*.

CTC (**connectionist temporal classification**) turns those per-column guesses into a string and makes training possible without alignments. Decoding merges repeated characters and then deletes blanks. Training sums the probability of *every* column labelling that would collapse to the correct string, so the network is free to place characters wherever it likes.

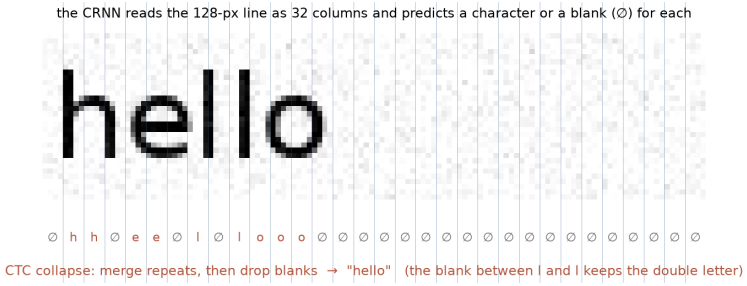


Figure 9: CTC on a rendered line

A 128-pixel line read as 32 columns, with one plausible labelling. Merging repeats and dropping blanks gives "hello". The blank between the two l's is what stops them merging into one. Real output would put blanks and characters wherever the network found the evidence.

The math

CTC marginalizes the alignment away. The network predicts a distribution over the alphabet plus a *blank* at every image column, and the loss sums the probability of every column labelling that collapses to the target:

$$p(y \mid x) = \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t} p(\pi_t \mid x),$$

where $\mathcal{B}$ deletes blanks and merges repeats. The blank is what makes repeated characters expressible at all — "AA" needs a blank between the two A's, or it collapses to "A" — and the sum is computed by a forward-backward recursion rather than enumerated.

Greedy decoding is not the most likely string. It takes the most likely *alignment* and collapses it, and those differ whenever one string's probability is spread over several alignments. Prefix beam search fixes that by scoring strings, which requires tracking two numbers per prefix — the probability it ends in a blank and the probability it ends in its final symbol — because which applies decides whether the next repeated character extends the prefix or merges into it. A test pins the case where they disagree: with two columns each splitting mass 0.55/0.45 between blank and 'A', greedy returns `""` and the beam returns `"A"`.

CER is an edit distance, not a per-position mismatch, because a model that drops one character would otherwise score as if everything after it were wrong too. It is pooled over the corpus rather than averaged per line, and it is not bounded by 1 — a hallucinating model can insert more than the reference holds.

What we measure, and why

- **CER (character error rate)** — the minimum number of character insertions, deletions and substitutions needed to turn the prediction into the truth (the **edit distance**), divided by the truth's length and pooled over the test set. It is used instead of per-position matching because one dropped character would otherwise make every later position "wrong". It can exceed 1 if the model hallucinates.
- **WER (word error rate)** — the same edit distance counted in whole words. One wrong letter spoils a word, so WER is always the harsher of the two, and the gap between them shows whether errors are scattered or concentrated.
- **CER on held-out fonts** — typefaces never seen in training, which tests generalisation rather than memorised glyph shapes. The rung found that its "held-out" split did not hold out what it claimed.
- **Greedy vs beam-search decoding** — whether picking the best column labelling and picking the best string disagree in practice.

What I built

A CRNN: convolutions pooling height to 1 while pooling width only twice, so a 32×128 image becomes 32 columns, then a two-layer BiLSTM and a per-column head. The width ratio carries a hard constraint — CTC cannot emit more characters than it has columns, and needs a spare column between repeats — so the rung checks it and exits loudly rather than discovering it as a silent `inf`.

Results

Setup: 6000 train lines, 12 epochs; medians over seeds 0–2.

split	greedy CER	beam-15 CER	WER
seen fonts	0.0000	0.0000	0.0033
held-out fonts	0.1636	0.1621	0.4900

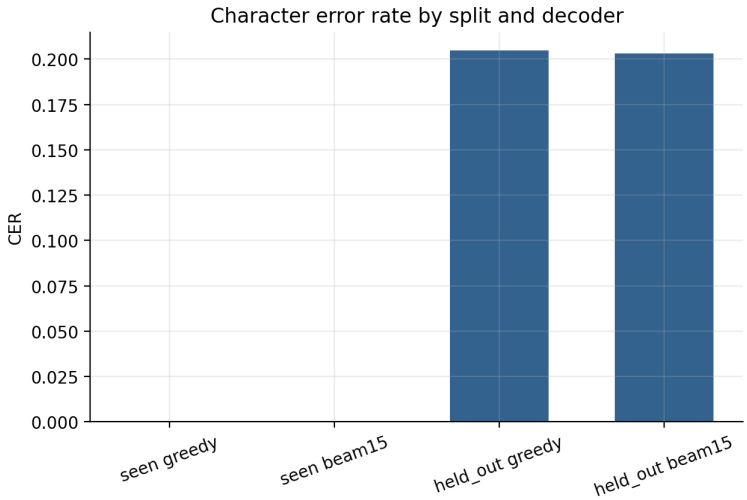


Figure 10: Decoders

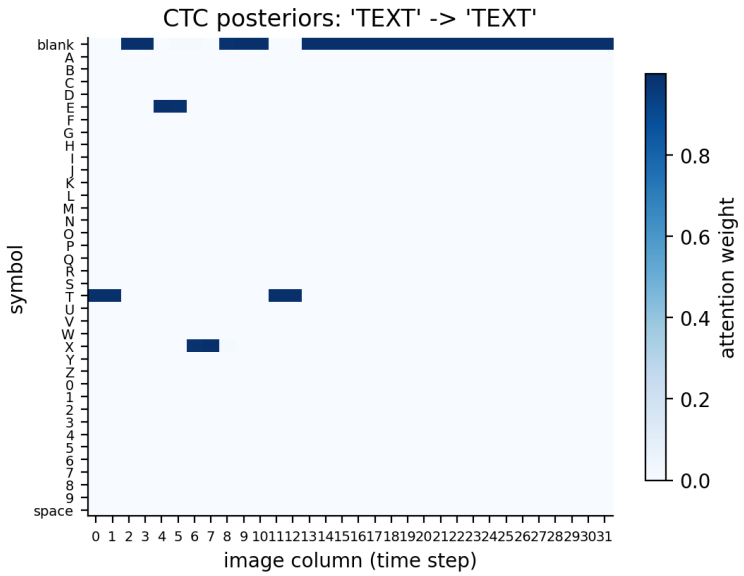


Figure 11: CTC posteriors

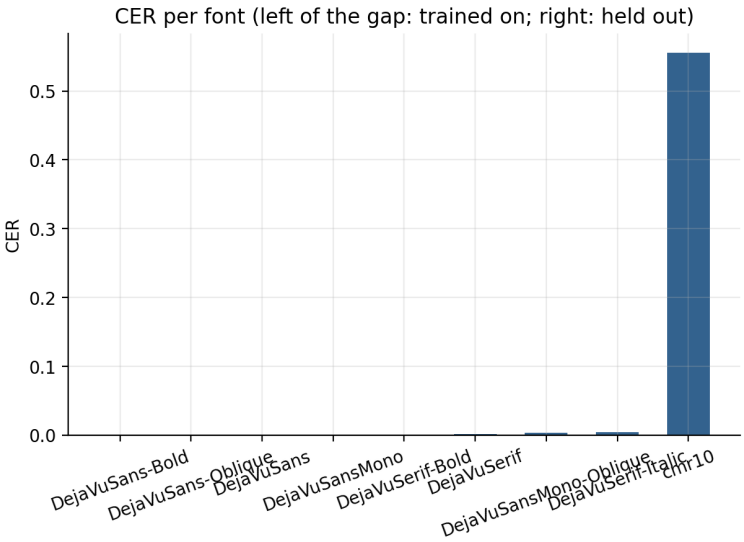


Figure 12: Error by font

What surprised me

Beam search buys **nothing**: 0.0000 CER at median against greedy, while held-out fonts cost 0.164. The textbook decoder upgrade, implemented properly with separate blank and symbol mass, is worth zero here, and the entire error budget is generalization. Decoding was never the bottleneck.

And the part-H finding I am least comfortable with. Breaking the held-out error down per font:

held-out font	CER
DejaVuSansMono-Oblique	0.0038
DejaVuSerif-Italic	0.0052
cmr10 (Computer Modern)	**0.4666**

The "font gap" is one typeface. Two of my three held-out faces are read at 0.4–0.5% CER — essentially perfectly — because they are *DejaVu family members* whose siblings (DejaVuSansMono, DejaVuSerif) are in the training set. Only Computer Modern, a genuinely different design, is hard, and it carries the entire aggregate gap on its own.

So the honest statement is not "the model fails on unseen fonts" but "the model generalizes across weight and slant within a family and fails on an unfamiliar letterform" — which is a different and more useful claim. My held-out split was badly designed: I picked three faces for variety without checking whether they were near-duplicates of training faces, and the aggregate number hid that until the per-font breakdown went in. Every statement I made about this gap before the breakdown existed was an average over two easy cases and one hard one.

Open threads

A held-out set chosen by typeface family rather than by file, which is what would make the gap measure what I claimed. Font augmentation during training. And synthetic degradation — blur, skew, JPEG — which is where real OCR error comes from and which this data has none of.

Further reading

CTC; CRNN; What Is Wrong With Scene Text Recognition Model Comparisons?.

H4 — Two ways to draw the image-to-string interface

Question. CTC assumes the output is monotonic in the input and conditionally independent given the columns. An attention decoder assumes neither and gains a language model over its own output. What does each assumption buy and cost?

The idea

There are two standard ways to connect an image to a string:

- **CTC** (H3): one prediction per image column, collapsed afterwards. It assumes the text runs left to right, one column after another (*monotonic*), and it predicts each column independently, so it cannot use knowledge of spelling.

- **Attention decoder**: generate the string one character at a time, like the C2 decoder. At each step, look back over all image columns with attention and decide the next character *given the characters already written*. That gives it an implicit language model, but nothing forces it to move steadily left to right, so it can lose its place, repeat itself or stop early.

A third arm, **TrOCR**, is a large pretrained transformer OCR model used zero-shot, for scale.

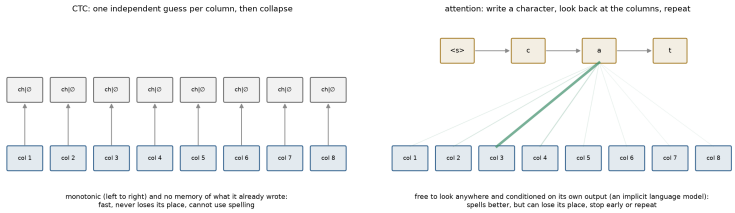


Figure 13: CTC and attention decoding

Left: CTC labels every column independently. Right: the attention decoder writes "a" by attending mostly to column 3, then moves on conditioned on "c", "a". Same encoder in both arms here, so the comparison isolates the decoder.

The math

The independence assumption is the real difference, and it is visible in the factorization each model assumes:

$$p_{\text{CTC}}(\pi \mid x) = \prod_t p(\pi_t \mid x) \qquad \text{against} \qquad p_{\text{AR}}(y \mid x) = \prod_t p(y_t \mid y_{<t}, x).$$

CTC's columns are conditionally independent given the image, so it cannot condition on what it already wrote — it cannot spell. The autoregressive product carries $y_{<t}$, which is an implicit language model over the label distribution, obtained for free and never asked for.

The monotonicity assumption is the other. CTC's collapse is order-preserving by construction, so alignment is never something the model can get wrong. An attention decoder chooses where to look at every step and must *learn* both the alignment and when to stop — and an emitted EOS is a decision, not a structural consequence.

Attention is C2's, pointed at image columns rather than source tokens: $\alpha = \text{softmax}(h_t^\top H/\sqrt{d})$ with context $\sum_i \alpha_i H_i$, plus input feeding so the decoder knows where it just looked.

What we measure, and why

- **CER** on lines of the lengths seen in training (one and two words).
- **CER on longer lines (three words) than any seen in training** — a direct test of the monotonicity assumption: CTC has no notion of "how far along am I", while the attention decoder must learn when to stop.
- **The two-words-on-a-wider-canvas control** — a third word on the same 128-px canvas also shrinks the characters, so a failure could come from resolution rather than length. This control separates the two.
- **Held-out WER and exact-line accuracy** — alongside CER, because an attention decoder with a language model can be worse per character and better per word, and it is here.
- **Font gap, length gap and width gap** — held-out CER minus seen CER along each axis of generalisation, so a single number says *which* kind of novelty hurts which decoder.

What I built

Three arms on byte-identical data. CTC is H3's CRNN, imported rather than reimplemented. The attention arm uses the *same* convolutional encoder and the *same* dot-product attention, so the comparison isolates the decoder. TrOCR is zero-shot, never trained here.

The experiment that matters is length: both arms train on one and two words and are tested on three. Critically there is a **two-words-on-the-wider-canvas control**, because at a fixed 128px a third word also shrinks the glyphs — and without separating those, a resolution failure reads as a length failure.

Results

Setup: medians over seeds 0–2; 0.4M / 0.9M / 62M parameters.

	CTC	attention	TrOCR
seen CER	0.0002	0.0000	0.0111
held-out CER	**0.1756**	0.1956	**0.0101**
held-out WER	0.4094	**0.2596**	0.0259
held-out exact	0.6125	**0.7013**	—
font gap	0.175	0.196	**−0.001**
length gap	**0.0003**	0.2202	—
width gap (control)	**−0.0002**	0.1393	—

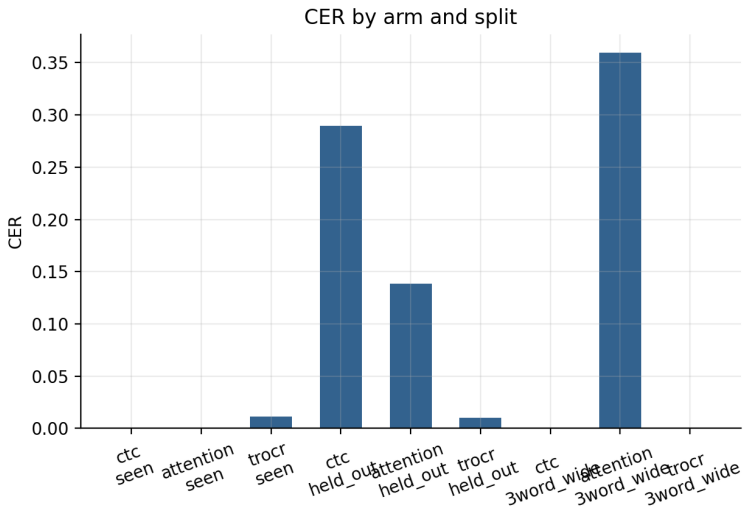


Figure 14: Arms

What surprised me

Three things.

CER and WER disagree about which decoder is better, on the same run. CTC wins on CER (0.176 vs 0.196); attention wins decisively on WER (0.260 vs 0.409) and exact match (0.701 vs 0.613). The CER→WER ratio is 2.5 for CTC and 1.3 for attention: the autoregressive decoder's implicit language model **concentrates** errors into fewer words, while CTC scatters independent character errors across many. Report CER and you ship CTC; report exact match and you ship attention. Nothing about the models differs between those two sentences.

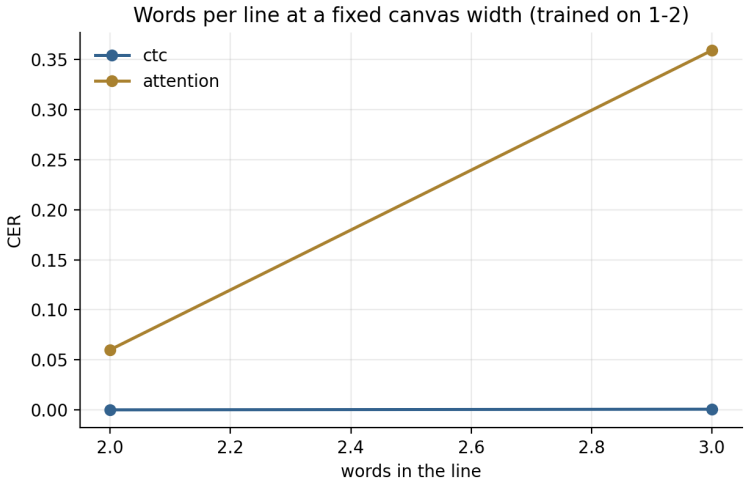


Figure 15: Length generalization

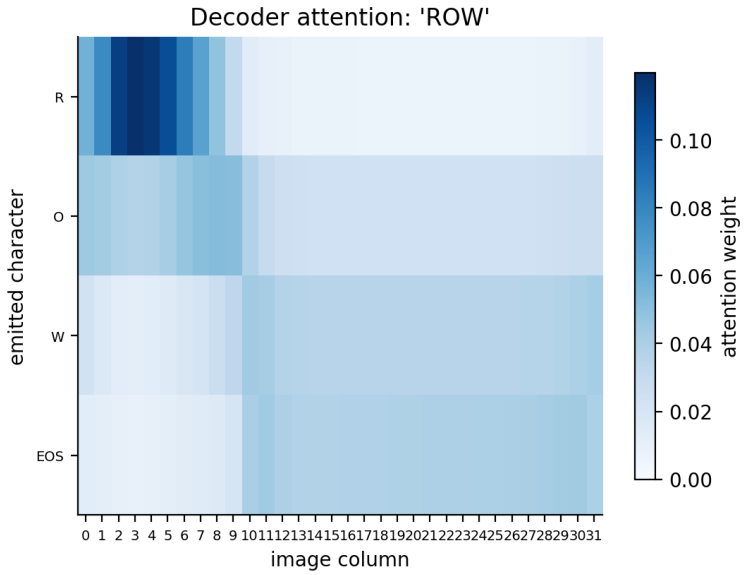


Figure 16: Attention map

The control did most of the work. The attention decoder's length gap is +0.220 — but **+0.139 of it is the wider canvas alone**, before a third word is involved. Only +0.081 is the unseen length. Had I run this without the control I would have written "attention fails to generalize to unseen length, +0.22 CER" and been 63% wrong about the cause. CTC meanwhile is *exactly* invariant to both: length gap +0.0003, width gap −0.0002. Its length bias at three words is −5.44 characters, i.e. it truncates, and it never gets a three-word line fully right.

TrOCR's font gap is negative. −0.001: it reads the held-out faces marginally *better* than the trained-on ones. Combined with H3's per-font breakdown, this settles the question — Computer Modern is not intrinsically hard, it was unfamiliar to a model trained on six faces. A model trained on real printed text has never heard of the problem.

Open threads

Beam search or a language model on the attention arm, which is where its implicit LM could be made explicit. Scheduled sampling for the exposure bias behind the truncation. And the obvious missing arm: fine-tuning TrOCR on this data rather than only reading it zero-shot.

Further reading

TrOCR; Attention Is All You Need; Baek et al. on comparing recognizers honestly.

H5 — The pipeline, and where its error actually lives

Question. Detect then recognize is how every classical OCR system works, and it introduces a failure neither stage can exhibit alone: the second consumes the first's mistakes. Whose fault is the error?

The idea

A classical OCR system is a **pipeline** of two models: a **detector** (H1) finds the text lines on a page, the lines are cropped out, and a **recognizer** (H3) reads each crop. Each stage can be trained and tested on its own, but the pipeline has a failure mode neither stage shows alone: a box a few pixels too narrow clips a character, and the recognizer, however good, cannot read what it was not given.

This rung builds that pipeline on synthetic pages whose ground truth is exact, and then *attributes* the pipeline's error to its stages: it scores the same recognizer on the true crops and on the detector's crops, and the difference is the detector's fault.

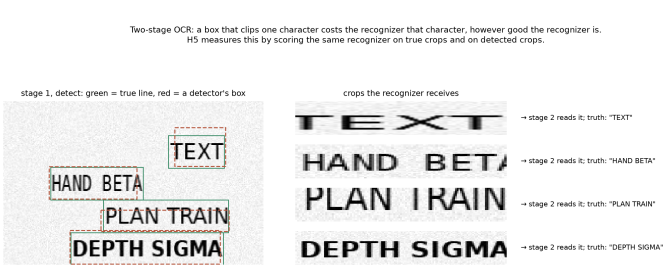


Figure 17: A two-stage OCR pipeline

A real page from the rung's generator. Green boxes are the true lines, red dashed boxes an imperfect detector's output (jittered here for illustration). The crops on the right are what the recognizer actually receives, and one of them has already lost the end of "BETA".

The math

A single end-to-end number cannot answer that, so the same recognizer is measured three ways: on ground-truth crops (detection assumed perfect), on detected crops (the pipeline as deployed), and on ground-truth crops with each box edge jittered by a known amount (detection error simulated, detector removed from the loop). The third turns the gap between the first two from an observation into an explanation.

The jitter is per edge, not a shift. A uniform translation moves the crop but keeps the text framed inside it; independent edges change the *framing* — clipping a first letter, admitting a neighbour's descender — which is the failure a real detector produces.

The end-to-end metric is page-level CER over lines joined in reading order:

$$\mathrm{CER}_{\mathrm{page}} = \frac{\sum_p d\Big(\bigoplus_{i\in\sigma(p)} \hat{y}_i,\ \bigoplus_{i\in\sigma^\star(p)} y_i\Big)}{\sum_p \Big|\bigoplus_{i\in\sigma^\star(p)} y_i\Big|},$$

with $\oplus$ concatenation and σ the reading order induced by the predicted boxes. Writing it this way makes the choice explicit: because the prediction is a concatenation over *detected* lines and the reference over *true* ones, a missed line becomes a run of deletions, a spurious one a run of insertions, and a wrong σ costs the edit distance between two orderings. A per-line average would score only the lines that were found — exactly the population a broken detector biases.

What we measure, and why

- **End-to-end page CER** — edit distance between everything the pipeline read on a page and the page's true text. This is what a user of the system experiences.
- **Recognizer CER on ground-truth crops** — the recognizer's own error with a perfect detector.
- **Error attribution** — the share of end-to-end error that disappears when detected crops are replaced by true ones. It is the rung's key number, and it turns "the pipeline is bad" into "this stage is responsible".
- **Detector recall and IoU of matched boxes** — the detector's own report card.
- **The jitter-augmented recognizer** — trained on deliberately misaligned crops, to measure whether the diagnosed problem has a cheap cure.

What I built

H1's detector — its grid arithmetic generalized from square scenes to 160×256 pages, so H5 inherits H1's round-trip test rather than growing a private copy — feeding H3's CRNN, trained on ground-truth crops. Training the recognizer on clean crops and testing it on detected ones is the point: that distribution shift *is* the detector's error. Plus a second recognizer trained with $\pm$4px jittered copies, because a diagnosis implies a remedy and the remedy should be measured.

Results

Setup: 2500 train / 500 test pages, medians over seeds 0–2.

detector: recall 0.854 [0.611, 0.929] · mAP@0.5 0.730 [0.359, 0.807] · matched IoU 0.671 [0.523, 0.731] · 1.18 spurious boxes per page

	clean recognizer	+ crop augmentation
arm 1 — ground-truth crops	0.0001	0.0000
arm 2 — full pipeline	**0.5121**	**0.3670**
matched lines only	0.3640	0.1868
tolerance (5% CER)	±0.33 px	**±7.78 px**

detection's share of page error	**0.9998** [0.9996, 1.0000]

Jitter curves (line CER at 0/2/4/8/16 px) — clean: 0.0001, 0.305, 0.505, 0.709, 0.863. Augmented: 0.0000, 0.0001, 0.0003, 0.053, 0.427.

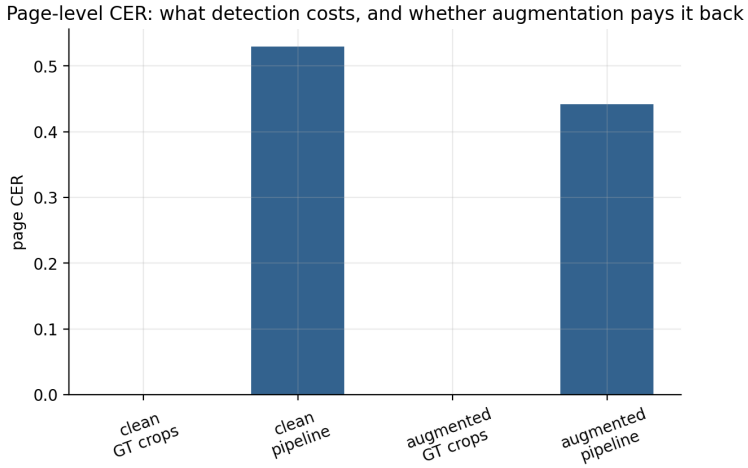


Figure 18: Attribution

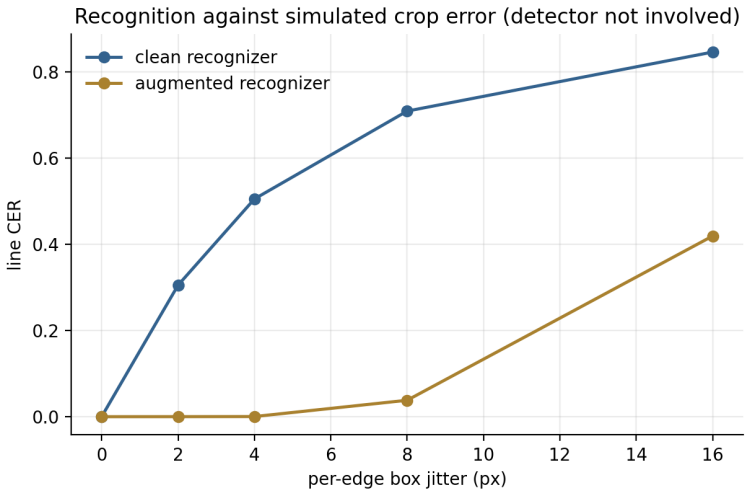


Figure 19: Jitter

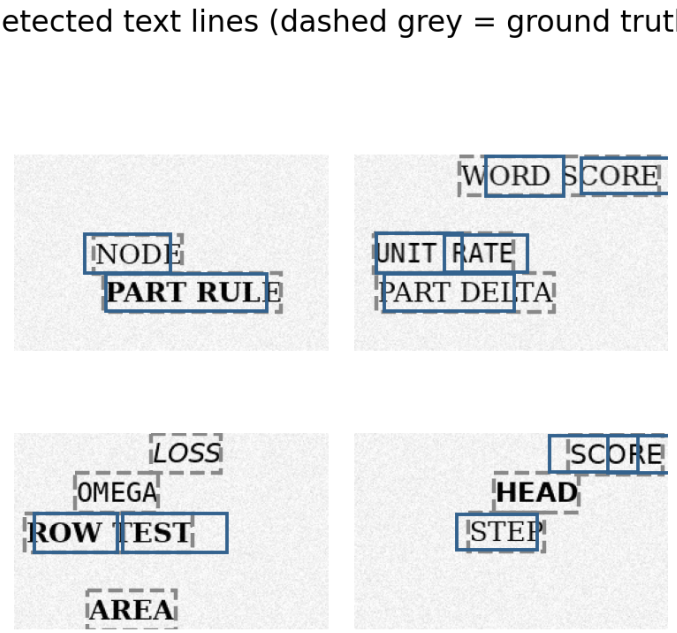


Figure 20: Pages

What surprised me

Detection is 99.98% of the error on every seed. The recognizer on ground-truth crops is essentially perfect (0.0001 page CER) and the entire 0.512 appears the moment it is handed real boxes. A pipeline's quality is its weakest interface, not its weakest model.

Arm 3 says why, and the number is brutal: interpolated, the clean recognizer crosses 5% CER at **0.33 pixels** of per-edge crop error. It was trained only on exact crops, has never seen a misframed line, and is brittle to a degree that makes the detector's 0.671 matched IoU fatal rather than merely mediocre.

The remedy worked, and mostly did not help. Training on jittered crops made the recognizer **24× more tolerant** ($\pm 0.33 \rightarrow \pm 7.78$ px) at *negative* cost on clean crops, and halved error on lines the detector found ($0.364 \rightarrow 0.187$). And the pipeline still sits at 0.367. Augmentation fixes framing; it cannot read a line that was never cropped. With recall at 0.854, the deletions from missed lines dominate everything crop quality controls. I had framed brittleness as *the* explanation for the pipeline gap — it is real, dramatic, and worth about a sixth of the total.

A single-seed number I reported and had to withdraw. The first H5 run gave recall 0.611 and mAP 0.359, and I described the detector's weakness using those. Across three seeds they are $[0.611, 0.929]$ and $[0.359, 0.807]$: I had quoted the *worst* seed as typical. The attribution conclusion is unaffected — 99.98% on all three — but the magnitude of the detector's failure is a range, not a point.

Open threads

Anchors with aspect priors: text lines are 54–189px wide and 32px tall, extreme ratios sharing one log-size regression, on a 20×32 grid that is 99.6% negative cells. Recall is the term that matters, so a lower score threshold and a recall-oriented detector would move the pipeline more than any recognizer change. And joint training through the crop, which is the last thing you can do before giving up on stages entirely.

Further reading

EAST; CRAFT; PP-OCR on what a practical pipeline contains.

H6 — Deleting the interface

Question. H5 measured that the stage boundary holds 99.98% of the error. A vision-language OCR model removes the boundary rather than improving either stage. What does that cost?

The idea

Modern OCR increasingly removes the pipeline entirely. A **vision-language model** (**VLM**) reads the whole page image and *writes out* the text as a language model would, with no detector, no crops and no hand-built interface between stages. The architecture has three parts:

1. a **vision encoder** (a ViT, C4) turns the page into a sequence of patch tokens;
2. a **projector** merges neighbouring patches and maps them into the language model's embedding space;
3. a **language-model decoder** (C3/C5) generates the transcription token by token, attending to the image tokens.

The price is size and compute. This rung prices it exactly for two real models, **PaddleOCR-VL** (0.9B parameters) and **Unlimited-OCR** (3.3B, a mixture-of-experts decoder), by parsing their published configurations. The section after this rung then runs them on H5's pages.

The architecture diagrams further down show both models' component stacks.

The math

There is no box, so there is no box error. A page becomes visual tokens and the transcript is decoded autoregressively from them; detection, NMS, cropping and reading order all cease to exist as separately-failing components. The architectural question is what replaces them.

Token count is the cost driver, because attention is quadratic:

$$\text{FLOPs}_{\text{attn}} \approx 4LHT^2 d_h, \qquad T = \frac{HW}{P^2 m^2}$$

for patch size P and a spatial merge factor m. PaddleOCR-VL's 2×2 merge concatenates neighbouring patches before the projector, dividing T by four and therefore attention by **sixteen** — the cheapest available lever on a quadratic.

The KV cache is $2LH_{kv}d_h$ bytes per token per dtype byte, which is where grouped-query attention pays: 16 query heads against 2 key/value heads makes the cache 8× smaller at unchanged query capacity.

Mixture-of-experts decouples parameters held from parameters run. With E routed experts of width d_{moe} and top-k routing plus s shared experts, a layer holds $(E + s)$ experts' worth and runs $(k + s)$.

What we measure, and why

- **Parameters per component** (vision, projector, text), validated first on a model the rung can load and count (TrOCR, 1.3% error), so the arithmetic is checked before it is trusted.
- **Held vs active parameters** for the mixture-of-experts decoder — what must sit in memory against what each token actually uses.
- **Image tokens and attention FLOPs by page size** — the cost driver. A full A4 page at 300 dpi becomes 11,214 tokens, so page resolution, not parameters, often decides whether a model is practical.
- **KV cache per token** — the decode-time memory cost (C5).

What I built

An accounting rung, not a benchmark. PaddleOCR-VL's architecture is **parsed out of candle's `config.rs` at run time** rather than quoted, so a fork that moves is detected instead of silently misdescribed; the 17 parsed defaults independently match the published `config.json`. The parameter formulas are validated against a model the rung can actually load and count — TrOCR, predicted 60.8M against measured 61.6M, **1.3% off** — so the budgets below are a checked calculation rather than an assertion.

Results:

component	parameters
PaddleOCR-VL vision (NaViT, 27 layers, 1152 wide)	411.4M
PaddleOCR-VL projector (2×2 merge + MLP)	26.0M
PaddleOCR-VL text (ERNIE-4.5, 18 layers, GQA 16:2)	466.7M
PaddleOCR-VL total	**904.0M** (45.5% vision)
Unlimited-OCR SAM-ViT-B tower	85.0M
Unlimited-OCR CLIP-L tower	302.3M
Unlimited-OCR text (MoE, 64+2 experts, top-6)	**2.93B held / 740M active**

page	patches	tokens after 2×2 merge	attention
384px	784	196	4.8 GFLOP
1024px	5,476	1,369	233.2 GFLOP
A4 at 300dpi	44,678	11,214	—

KV cache 18.0 KiB/token, 8× less than multi-head would need.

What surprised me

The vision tower is **45.5%** of PaddleOCR-VL — nearly as large as the language model it feeds. I had assumed the encoder would be a thin front-end on a big decoder, the way TrOCR's DeiT-small is. For document understanding the reading *is* the hard part, and the parameter split says so.

The exchange rate against H5 is **1,853×** — 904M parameters against 488k — for a model that also handles 109 languages, tables and formulas, so the comparison is not clean. But the structural

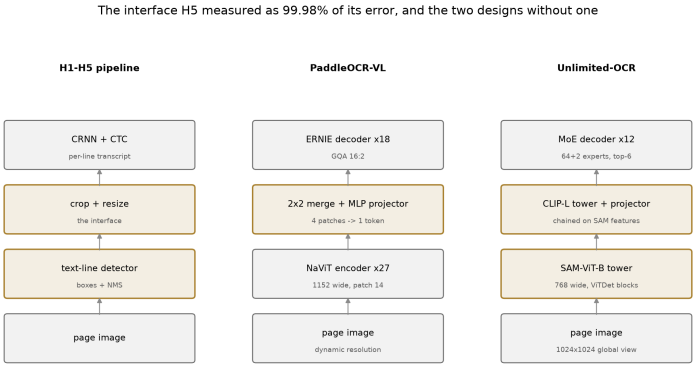


Figure 21: Architectures

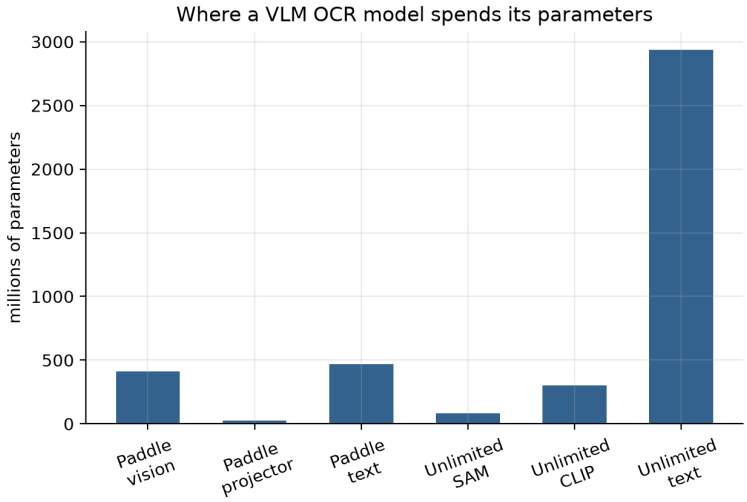


Figure 22: Parameter budget

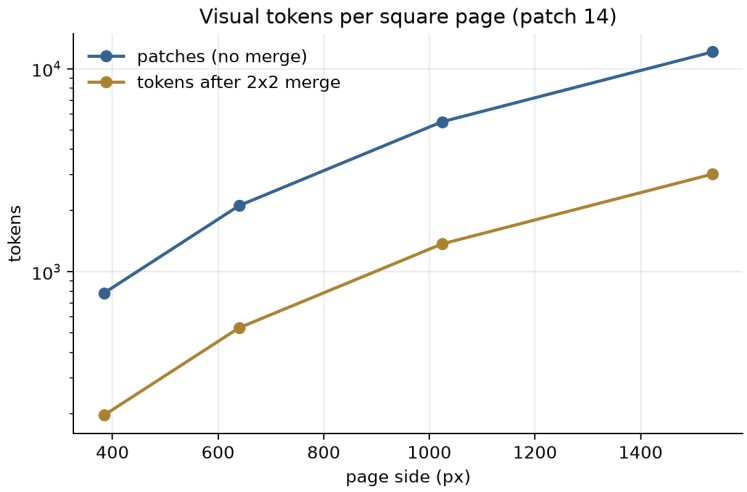


Figure 23: Token cost

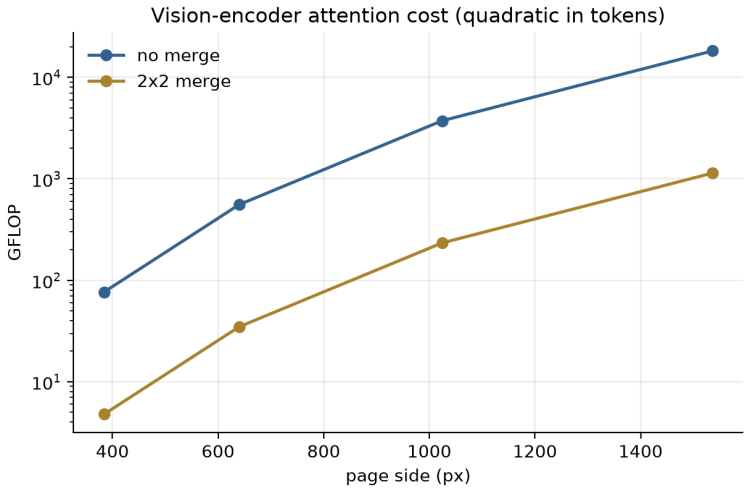


Figure 24: Attention cost

point survives it: every component H5 measured as its error source is absent from that 904M, not improved within it.

And an error I made worth recording. I first counted Unlimited-OCR's text tower as a dense stack, which understated it by 4× and erased the distinction MoE exists to create. I had also *guessed* its `intermediate_size` at 6848 because candle's config carries no defaults for it — the guess happened to be right, which is worse than being wrong, because it would have shipped ungrounded and looked correct. Both are now read from the published config with provenance recorded beside them.

Open threads

The head-to-head this rung deliberately did not attempt has since been run: PaddleOCR-VL and Unlimited-OCR on H5's own pages, in the models, actually run below, which also corrected this rung's token formula. Still open: measuring the vision tower's activation memory, which at 11,214 tokens for an A4 scan is likely the real deployment constraint rather than parameters. And DeepSeek-OCR's compression claims, which is the lineage Unlimited-OCR comes from.

Further reading

PaddleOCR-VL; NaViT; Donut; GQA.

The models, actually run

Everything above H6 is a model I trained; H6 itself is arithmetic over two published configs. Both of those are ways of being wrong that the rung cannot detect on its own — so the models H6 prices were downloaded and run on H5's own test pages, against H5's own ground truth, scored with H3's own CER.

The pages are written by `tools/real_models/write_pages.py`, which replays the same rng stream `h5_document_pipeline` uses: 2500 train pages drawn first, so page k here is page k of H5's test split. Each page is saved twice — at its native 256x160 and upscaled 4x with Lanczos. **Upscaling adds no information**, so the ground truth is untouched; it exists only to remove the

excuse that the page is smaller than the model's patch grid, and it turns out to be the variable that matters.

Results

Setup: 10 pages, 26 lines; raw values in `data/h-real-models.json`.

model	params	page	CER	WER	s/page
H5's pipeline (clean arm)	0.49M	native	0.512	—	—
H5's pipeline (augmented arm)	0.49M	native	0.367	—	—
H5's recognizer on gold crops	0.42M	crops	0.0001	—	—
PaddleOCR-VL (T4, fp16)	0.96B	4x	**0.000**	**0.000**	3.33
PaddleOCR-VL (T4, fp16)	0.96B	native	0.039	0.032	0.83
PaddleOCR-VL (CPU, fp32)	0.96B	4x	**0.000**	**0.000**	96.2
Unlimited-OCR (T4, bf16)	3.34B	4x	**0.000**	**0.000**	7.18
Unlimited-OCR (T4, bf16)	3.34B	native	0.317	0.320	3.10

Both models transcribe every line of every page **exactly** at 4x. H5's pipeline, trained on this exact distribution, reads the same pages at 0.512 CER — and the gap is not a recognition gap, because H5's own recognizer reads gold crops at 0.0001. It is the interface, which is what H5 concluded when it attributed 99.98% of the pipeline's error to detection, and what H6 argued was worth deleting. Here is that argument as a measurement rather than an argument: deleting the interface costs **2,000x the parameters** — 0.96B against 0.49M — and takes the page CER from 0.512 to 0.000.

What surprised me

Resolution is the whole story for the smaller model, and a cliff for the larger one. PaddleOCR-VL degrades gracefully — 0.000 to 0.039 CER as the page shrinks 4x. Unlimited-OCR, three and a half times the size, goes from 0.000 to 0.317, and the failure mode is not misreading: on several native-resolution pages it returns **an empty transcript** rather than a wrong one. Its refusal is better calibrated than its reading, which is a design choice I would not have predicted from either config, and a reminder that "bigger model" and "more robust" are different claims.

Unlimited-OCR does detection and recognition in one token stream. It emits `<|det|>text [687, 48, 984, 163]<|/det|>SCORE` — a grounded box immediately before the line's text. H5's interface is not *gone* in this model; it has been moved inside the sequence, where the two stages share gradients instead of passing crops. That is a sharper statement of H6's point than H6 makes.

Cheap hardware is the difference between the two OCR models, not quality. PaddleOCR-VL reads these pages perfectly on four CPU cores — 96 s/page, but perfectly, and it is 29x faster on a T4 (3.33 s/page). Unlimited-OCR needs the GPU. On this evidence the 0.96B model is strictly the better deployment for this document class.

A correction to H6, which is why this was worth doing. H6's `page_tokens` floored the patch grid. A NaViT-style encoder *pads* a page to whole patches rather than cropping the remainder, so the count is a ceiling:

$$\text{tokens} = \left\lceil \frac{H}{p \cdot m} \right\rceil \cdot \left\lceil \frac{W}{p \cdot m} \right\rceil .$$

For the 1024x640 page that is 23 x 37 = **851**, which is exactly what PaddleOCR-VL's processor emitted. The shipped formula said 821. The old answer is now pinned in a test so it cannot come back.

The second half of the check failed and stayed failed, which is the more useful half. At native size the geometry says 60 tokens; the processor emitted **198**, 3.3x more, because it will not shrink a page below a floor. That is a preprocessing policy, not something derivable from the model config, so it is recorded as a measured constant (`MEASURED_TOKEN_FLOOR`) rather than folded into the formula as though it had been derived. **A small page does not cost little.**

The decoders got the same treatment, and it belongs with C5 rather than here: DeepSeek-V2-Lite, Gemma-4 and both Qwen 27Bs are in part C's writeup with `data/c-real-models.json`.

What did not run, and why. `Qwen3.8-Flash-Next` is 360GB of weights and is out of reach here at any quantization. Everything else asked for did run, and is reported with part C.

The same model through candle, and what two implementations agreeing buys

Everything above ran under transformers. The accompanying candle fork implements PaddleOCR-VL directly, so the same weights were run through it on the same T4, on the same pages, scored by the same function:

runtime	4x CER	native CER	native WER	s/page (4x)
transformers, fp16	0.0000	2/51 = 0.0392	1/31 = 0.0323	3.33
candle fork, f16	0.0000	2/51 = 0.0392	1/31 = 0.0323	4.68

Identical to sixteen digits (both runtimes produce the same exact fractions, 2/51 and 1/31), and they fail on the same page for the same reason: page 5's two lines are both `SIGMA`, and both runtimes emit one. So the native-resolution error is not a misreading at all — it is a **repeated-line collapse**, which I would not have diagnosed from one implementation, because one implementation's quirk and a model's behaviour look the same from inside it.

candle also prints its own geometry: `1024x640 -> 1036x644 (46 x 74 patches)`. That is 3,404 patches and **851 tokens** after the 2x2 merge — the corrected ceiling above, confirmed by an implementation that has never seen my arithmetic.

Building it. `CUDA_COMPUTE_CAP=75 cargo build --release --features cuda --example paddleocr-vl`: **18m 44s**, and **37 seconds** to add the `gemma4` and `deepseekv2` examples afterwards. The cost is the first build, not the model — I had guessed "30+ minutes" for the build and used that guess as a reason to reach for transformers instead, which was wrong twice over: wrong about the number, and wrong to let an unmeasured estimate decide the approach.

Two things had to be fixed to get there, both environmental rather than architectural:

- `candle-kernels/src/compatibility.cuh` defines `__hmax_nan`/`__hmin_nan` under `#if __CUDA_ARCH__ < 800`, but **CUDA 12.8 already defines them for arch >= 530**, so `nvcc` fails on `src/affine.cu` with a redefinition. Adding `&& CUDART_VERSION < 12000` to the guard fixes it.
- `--bf16` on an sm_75 card dies with `DriverError(CUDA_ERROR_NOT_FOUND, "named symbol not found")`, because candle's bf16 kernels need sm_80. Dropping the flag works — and the error names a missing symbol rather than the missing capability, which cost me a diagnosis.

And one that is architectural. `ignite-perf`'s `qwen3_5`, `gemma4` and `unlimited_ocr` call `candle_nn::fused::static_decode`, which is behind `#[cfg(feature = "cuda")]`, from call sites that are not — so a build *without* `--features cuda` fails with 51 errors in `candle-transformers`. A CPU-only build of those models is impossible on this branch; gating the call sites the same way would fix it.

Unlimited-OCR, and two bugs that only exist above a batch of one

The fork implements Unlimited-OCR but ships no example for it. The catalog's `candle-ocr` executor does drive it, so this repo now carries that path with the ignite SDK taken out — hub loading instead of the asset loader, the same 1024-square mean-gray pad, the same grid search, the same sliding-window n-gram suppression, plain KV-cache decode instead of static R-SWA.

Getting it to run found two bugs, and they share a shape:

- `segment_anything/image_encoder.rs` adds its learned positional grid with `+`, which in candle requires equal shapes. The grid is `[1, h, w, c]` and a batch of crops is `[n, h, w, c]`, so the tiled path dies with `shape mismatch in add, lhs: [8, 40, 40, 768], rhs: [1, 40, 40, 768]`.
- With that fixed, `forward_crops` cats a transposed tensor and hands the result to a batched matmul, which rejects the strides.

Neither exists at a batch of one, which is exactly the path the executor runs — its comment says so: *"V1 scope: single 1024x1024 global view."* Code that is only ever called one way is only ever tested one way, and the untested way was two lines from working.

Results

Setup: 10 pages, f16, T4.

preprocessing	image tokens	CER	WER	empty	s/page
global view only	273	**0.229**	0.241	1/10	0.75
tiled, 5x3 grid	1,803	1.113	1.095	4/10	0.89
transformers `infer()`, crop_mode	—	**0.000**	0.000	0/10	7.18

Eight of ten pages are read exactly through the global view. The tiled path, once unblocked, is **worse at 6.6x the image tokens** — so whatever bounds the reference puts on its grid, this port does not have them, and a 1024x640 page should not be asking for a 5x3 grid. That is a preprocessing gap, not a runtime one, and PaddleOCR-VL is the control that proves the distinction: the same two runtimes agree to sixteen digits when the preprocessing matches.

The decoder attempts through the fork are with part C.

Reproduce: `tools/real_models/` — the scripts, and a README explaining the transformers version pin each checkpoint needs and why.

What part H settled

1. **A pipeline's quality is its weakest interface, not its weakest model.** H5's recognizer is perfect on clean crops and the pipeline is at 0.512 page CER, with 99.98% of that attributable to detection on every seed. Improving the recognizer 24× in robustness — which crop augmentation genuinely did, for free — recovered a sixth of it. The rest is lines that were never found.
2. **Detection has no metric until you choose one, and the choice is worth more than most modelling decisions.** The same predictions score 0.955 or 0.650 depending only on what overlap counts as correct, and the NMS dial that moves mAP@0.5 by 0.109 moves mAP@0.75 by 0.032. Reporting a single mAP without its threshold conveys almost nothing.
3. **Which decoder is "better" depends on which metric you report.** On identical held-out data CTC wins CER and the attention decoder wins WER and exact match, because an autoregressive decoder's implicit language model concentrates errors into fewer words while CTC scatters them. Both statements are true of the same run.
4. **Averaging hides mechanisms; score where the mechanism acts.** Skip connections looked worth +0.013 until the same predictions were scored on a 2px boundary band, where they are worth +0.030. The effect was never absent — whole-image mIoU was averaging the place skips work together with the place they don't.
5. **Controls and breakdowns changed three conclusions that a headline number got wrong.** H4's canvas-width control showed 63% of an apparent length-generalization failure was resolution. H3's per-font breakdown showed the entire "held-out font gap" was one typeface, with the other two held-out faces read at 0.4% CER because they were family members of training faces. H1's localization investigation ruled out three plausible explanations and left me with an honest "I don't know" for the per-class ordering. In each case the aggregate was not wrong, it was uninformative in a way that looked informative.
6. **The single-seed habit keeps costing me.** H2's "Dice beats cross-entropy" and H5's "recall 0.611" were both reported before the sweep and both failed it — the first was noise over a 0.012 spread, the second was the worst of three seeds quoted as typical. Together with part B's `usable_lrs`, part D's `augmentation_gain` and H5's own degenerate tolerance metric, that is six occasions in this repo where a number that looked like a result was an artifact of how it was computed or how few times it was run. The arm-level measurements were right every time; only the summaries lied.
7. **The field's answer to H5 was to delete the interface, not fix it, and running the real models settles whether that works.** H6 prices the deletion at 904M parameters and 196 tokens per 384px page. PaddleOCR-VL (0.96B) and Unlimited-OCR (3.34B) then read H5's own test pages at **CER 0.000** where H5's pipeline reads them at 0.512 — 2,000x the parameters to remove the interface, and the error goes with it. Whether that trade is worth making is a deployment question; that it works is now measured rather than argued.
8. **Arithmetic over a config is a hypothesis, and this one was wrong.** H6's token count floored the patch grid where a NaViT encoder pads it; the corrected ceiling gives 851 tokens for a 1024x640 page, exactly what the processor emits. The second prediction stayed wrong and is more useful for it: a 256x160 page is 60 tokens of geometry and 198 in fact, because the processor will not shrink a page below a floor. Being able to run the model is what separated the bug from the policy, and neither was visible from the config alone.

Cost of the part: ~3,600 lines including tests, 78 test cases, and a three-seed sweep in about 24 minutes on a T4 (H1 14s, H2 42s, H3 116s, H4 114s, H5 194s, H6 1s per seed). H5 dominates because it trains a detector and two recognizers; H6 is arithmetic.

Figures regenerate with `uv run python tools/writeup_figures.py H`*; the numbers they are quoted against live in* `data/h-metrics.json`*. The part README is the short version of this document.*